



Panasonic's car sensor

Panasonic has reportedly developed a new image sensor technology that should improve night vision in vehicles. <http://digit.in/ImageCar>



Passenger carrying drone

Residents of Dubai are reportedly getting an autonomous quadcopter capable of carrying passengers this summer. <http://digit.in/taxiDrone>



R2D2 and C3PO have sparked countless minds in the world of robotics



A NEW ROBOTIC HOPE

In the 1970's, manufacturing processes were increasingly using robotic arms to increase efficiency or get dangerous things done. Both the PUMA (Programmable Universal Manipulation Arms) and the SCARA (Selective Compliant Articulated Robot Arm) were developed and extensively used in the 1970s.

But something significant happened in the world of robotics in this decade that would inspire scientist and engineers for decades to come – Star Wars was released. The movie, set in a universe where humans and robots coexisted in perfect harmony, inspired many to look at the next level of human-like robots.

FROM HUMANOID TO HUMANLIKE

In 1986, Honda began a research project to develop robots that could interact with humans. Their goal was a robot that “should coexist and cooperate with human beings, by doing what a person cannot do and by cultivating a new dimension in mobility to ultimately benefit society.” Although this project would take almost a decade

to bear fruit, its results would be highly significant in the coming years.

In 1989, a research paper changed the direction of rover research. So far, robots were being used for space exploration, but the direction of design to build one big multipurpose robot instead of multiple small ones, was advocated by this paper; it influenced the way we designed rovers.

The 90s were full of robotic advancements, as if decades of research and study was finally culminating into results. For example, the 8 legged walking robot Dante explored Mt. Erebus in Antarctica. Dante II, a more robust successor, subsequently explored Mt. Spurr in Alaska in 2004. The Cyberknife, a stereotactic radiosurgery robot released in 1994, could treat tumours non-invasively with an accuracy that was comparable to human doctors.

After almost a decade of research, Honda's P2, and then P3 were ready to be publicly demonstrated. The former was the first self regulating bipedal robot while the latter was Honda's first completely autonomous robot.



Honda's P2 would go on to be further refined



ASIMO



AND WE'RE THERE... ALMOST!

The dawn of the new millennium saw robotics reach everyday life and people like never before. The ASIMO, the result of Honda's decade and a half worth of robotics research, was introduced to the world and it was the most advanced robot of its kind at the time.

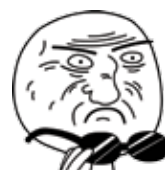
Weighing only 54kgs, ASIMO (Advanced Step in Innovative Mobility) was capable of running, walking, communication with humans, facial and environmental recogni-



AIBO

tion, voice and posture recognition, and interacting with its environment. Estimates in the same year showed that more than 700,000 robots were being used for industrial purposes around the world – more than half of them in Japan.

The AIBO also gained popularity around this time. Artificial Intelligence Robot, or AIBO, was a robotic dog capable of interacting with humans. While intended as a consumer market entertainment device, the AIBO had several remarkable capabilities like responding to over 100 voice commands and talking in a tonal language.



BEHOLD THOSE WHO INHERIT THE EARTH

To most of us, that statement might appear too exaggerated. While we are not claiming otherwise, it would be great if you could look around you and realise how much the world is already dependent on robots. These robots, by the way, are more sophisticated than they have ever been. The Roomba automatic vacuum cleaner has sold millions of units, showing how willing we are to welcome domestic robotic technology into our homes.

On the other side of the power spectrum, there are drones that can navigate the globe by themselves and perform dangerous aerial strikes while the human controller is seated in the safety of their office (or even asleep). On the other hand, artificial intelligence programs have beaten the world's best at their own games and even learnt how to bluff. 